

Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and shopping channels. They are interested in identifying which factors, such as discounts, reviews, seasons, and payment methods, influence customer purchasing decisions and repeat purchases.

The company wants to analyze customer behavior data to identify valuable customer segments, improve marketing strategies, and enhance customer engagement. By studying shopping patterns and consumer preferences, the organization aims to make data-driven business decisions and optimize overall customer experience.

You are tasked with analyzing the company's consumer shopping dataset to answer the following business question:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Deliverables

1. **Data Preparation & Cleaning**

Clean and transform the raw dataset by handling missing values, duplicates, formatting issues, and incorrect data types.

2. **Exploratory Data Analysis (EDA)**

Analyze customer purchasing patterns, product preferences, payment methods, and demographic trends.

3. **Data Analysis (SQL)**

Write SQL queries to extract insights related to customer segments, repeat purchases, shopping behavior, and purchase trends.

4. **Dashboard Development (Power BI)**

Build an interactive dashboard that visualizes customer behavior trends, spending patterns, category analysis, and customer insights.

5. **Presentation & Reporting**

Prepare a presentation summarizing project findings, customer insights, and business recommendations.

6. **GitHub Repository**

Upload all project files including datasets, SQL queries, dashboards, screenshots, and documentation in a properly organized GitHub repository.